Estimation: Bid Intelligence Platform

EMPA-ReMarkets — Bid Intelligence Platform **— Blazor Web App + .NET 10**

Scope Summary

Internal web application for REMarkets to replace spreadsheet/email-based bidding workflows. Covers bid management, inventory handling, allocation approvals, role-based access, act-on-behalf, audit trails, and order-ready exports. Authentication via Microsoft Entra. All data stored in clean, normalized SQL for future analytics consumption.

Architecture Summary

- Frontend + Backend: Blazor Web App (.NET 10) — unified C# codebase
- Render Modes: Static SSR for simple pages, Interactive Server for bid tables and allocation views
- UI Components: MudBlazor (free) with DataGrid for in-cell editing
- Database: Azure SQL Database with Entity Framework Core 10 (Code-First)
- Auth: Microsoft Entra ID via Microsoft.Identity.Web (built-in Blazor auth)
- Hosting: Azure App Service (3 environments: Dev, Test, Prod — TBD if already provisioned)
- Deploy: Single project, single deploy

Data Grid Library Options (In-Cell Editing — Core Feature)

The in-cell editing grid is the most critical UI component — it's where sales reps enter bids, quantities, and prices directly in a table (like Excel in the browser). Without this, every edit requires opening a separate form per line item.

Library	License	Cost	In-Cell Editing	Notes
MudBlazor DataGrid ★ Recommended	MIT (free, open source)	$0	★★★★ Good	Active community, good docs, evaluate in Sprint 0
Radzen DataGrid	MIT (free, open source)	$0	★★★★ Similar	Alternative if MudBlazor doesn't fit
AG Grid (Blazor wrapper)	Community: MIT / Enterprise: Commercial	~$1,100/dev	★★★★★ Best-in-class	Same AG Grid as React, has Blazor wrapper
Telerik UI for Blazor	Commercial (paid license)	~$1,000/dev/yr	★★★★★ Excellent	Full UI suite with professional support
Syncfusion Blazor	Community: free <$1M / Commercial	$0 or ~$995/dev/yr	★★★★ Very good	Free tier may apply

Library	License	Cost	In-Cell Editing	Notes
Manual (jQuery/JS)	N/A	$0	★★ Painful	Not recommended — conflicts with Blazor DOM

Recommendation: Start with MudBlazor (free, MIT). If during Sprint 0 the DataGrid doesn't handle the in-cell editing complexity needed, upgrade to Telerik or AG Grid (~$2-3K for 2-3 devs).

Why not build it manually? Blazor controls the entire page DOM through C#. Adding jQuery or raw JavaScript creates two systems fighting over the same page elements — guaranteed bugs.

MVP Definition

Core question: Can REMarkets run 100% of internal bid-to-allocation workflows through the platform instead of spreadsheets and email?

MVP goal: A working web application where sales reps can create offers, enter bids, upload inventory, and route allocations through an approval workflow — with full audit logging and role-based access.

MVP Scope (Must Have)

Without these, the project fails to replace the current spreadsheet/email process.

#	Module	Task	Complexity	Base Hrs	Factor	Adjusted	Notes
Project Setup							
1	Setup	Blazor Web App project scaffolding (Clean Architecture)	Low	12	×1.2	**14**	Single project, C# only
2	Setup	EF Core setup, initial migrations, seed data	Low	12	×1.2	**14**	
3	Setup	CI/CD pipeline (build + test + deploy)	Low	10	×1.2	**12**	Single project = simpler pipeline
Authentication & Authorization							
Option A: Azure AD Groups (Confirmed) — Roles managed in Azure AD — no user/role tables in app DB							
4	Auth	Entra integration — Microsoft.Identity.Web (built-in Blazor auth)	Medium	12	×1.5	**18**	Simpler than MSAL.js — built-in support
5	Auth	Azure AD Group setup + group claims mapping (5 groups: Sales Rep, Sales Manager, Admin, Finance, Executive)	Medium	10	×1.5	**15**	Configure groups in Azure AD, map to app roles via token claims
6	Auth	Dynamic role-based authorization (policies reading AD group claims) + app-layer enforcement	Medium	14	×1.5	**21**	API and services enforce roles from AD groups; app-layer enforcement required per client
7	Auth	AuthorizeView components (dynamic role-based UI rendering from AD claims)	Low	12	×1.2	**14**	Show/hide/disable UI elements based on group membership
Auth Option A subtotal — 68 hrs							
Customer Management							
8	Customers	Customer CRUD (API + Blazor pages)	Low	14	×1.2	**17**	Standard CRUD

#	Module	Task	Complexity	Base Hrs	Factor	Adjusted	Notes
9	Customers	Customer assignment to Sales Reps/Managers	Low	8	×1.2	**10**	
10	Customers	Customer deactivation (soft delete) + audit logging	Low	8	×1.2	**10**	
Inventory Management							
11	Inventory	Bulk CSV/Excel upload (parsing + validation + error handling)	High	24	×2.0	**48**	File parsing edge cases
12	Inventory	Inventory identity — composite uniqueness (PartNumber + condition + location + grade) + master vs source description logic	High	20	×2.0	**40**	Client-confirmed: net-new → populate master; existing → retain source only; master changes require Admin/Manager action
13	Inventory	Inventory adjustment records (each upload row = distinct transaction: batch ID, user, timestamp, qty, source file)	Medium	14	×1.5	**21**	Client-confirmed: never silently merge — full audit trail per upload row
14	Inventory	Inventory list view with state tracking (Available/Committed/Released)	Medium	14	×1.5	**21**	
15	Inventory	Inventory grouping into offers	Medium	12	×1.5	**18**	
16	Inventory	State transitions (automatic on approval + manual override)	Medium	16	×1.5	**24**	Business rules complexity
Bid & Offer Management							
17	Bids	Create/Edit offers (Blazor forms)	Medium	14	×1.5	**21**	
18	Bids	Offer list view with filtering and sorting	Medium	10	×1.5	**15**	
19	Bids	MudBlazor DataGrid — in-cell editing for line-item bids	High	28	×2.0	**56**	Core feature, DataGrid config
20	Bids	Bid uniqueness enforcement (one active bid per customer per line per offer) + revision tracking	Medium	10	×1.5	**15**	Client-confirmed: new revision on update, previous retained for audit, only latest active
21	Bids	Bid types support (bid/BIN/target pricing)	Medium	12	×1.5	**18**	
22	Bids	Winning/losing bid capture and status tracking	Medium	12	×1.5	**18**	
23	Bids	Bid history (view previous revisions per line item)	Medium	14	×1.5	**21**	
Allocation & Approval Workflow							
24	Allocation	Aggregated allocation view (all bids per offer)	Medium	14	×1.5	**21**	
25	Allocation	Admin override of winning customer	Medium	12	×1.5	**18**	
26	Allocation	Approval workflow (Pending → Approved/Rejected) — all manager-approved in Phase 1	High	24	×2.0	**48**	Client-confirmed: thresholds out of scope Phase 1; Admin has full authority including override
27	Allocation	Visual pricing indicators ("below floor" warnings) — thresholds configurable in future phase	Medium	8	×1.5	**12**	Client-confirmed: visual only in Phase 1, no auto-reject
28	Allocation	Automatic inventory commitment on approval	Medium	12	×1.5	**18**	
Allocation Reversal — Client-confirmed — must support in Phase 1							

#	Module	Task	Complexity	Base Hrs	Factor	Adjusted	Notes
29	Allocation	Allocation reversal (return inventory to Available, mark as Reversed, lock/regenerate export)	High	20	×2.0	**40**	State machine change + inventory rollback + export invalidation
30	Allocation	Reversal reason code (required) + full audit trail for reversals	Medium	8	×1.5	**12**	Reason code dropdown + audit entry
Allocation Tie-Breaker — Client-confirmed — system surfaces ties, manager resolves manually							
31	Allocation	Tie detection logic + resolution UI (surface tied bids, force manual winner selection)	Medium	8	×1.5	**12**	Admin/Manager picks winner from tied bids
Act-on-Behalf (Sales enters bids for customers) — Client-requested — needed from day 1							
32	Act-on-Behalf	Act-on-behalf capability — sales rep enters bids on behalf of a customer with full attribution	High	24	×2.0	**48**	Custom AuthenticationStateProvider; tracks who acted + for whom
33	Act-on-Behalf	Act-on-behalf UI (customer selector + visual indicator showing active customer context)	Medium	10	×1.5	**15**	Clear visual cue that rep is acting for customer
Split Allocations — Client-requested							
34	Allocation	Partial/split allocations across customers (divide quantity of a line item between multiple winners)	High	24	×2.0	**48**	Manual decision by manager, not automated
Audit & Compliance							
35	Audit	Audit logging middleware (insert-only AuditLog table)	Medium	16	×1.5	**24**	
36	Audit	Audit trail UI (view history per entity)	Medium	14	×1.5	**21**	
Order-Ready Exports							
37	Exports	CSV export from approved allocations	Low	8	×1.2	**10**	
Permissions UI							
38	Permissions	Role-based UI visibility (AuthorizeView across pages)	Medium	14	×1.5	**21**	
Testing & Bug Buffer							
39	QA	Unit tests (services + critical business logic)	Medium	20	×1.5	**30**	
40	QA	Integration tests (API/page endpoints)	Medium	14	×1.5	**21**	
41	QA	Bug fixing buffer (10% of MVP hours)	—	—	—	**90**	Based on ~900 hrs pre-buffer
MVP Total (Auth Option A)						**~990 hrs**	

Post-MVP Scope (Should Have)

Features that add value after MVP is stable.

#	Module	Task	Complexity	Base Hrs	Factor	Adjusted	Source
42	Allocation	Escalation paths (auto-escalate if not approved in X time)	Medium	16	×1.5	**24**	Suggested enhancement
43	Exports	PDF export (order-ready documents via QuestPDF)	Medium	14	×1.5	**21**	Client mentioned "order-ready exports" — CSV is in MVP, PDF is extra
44	Audit	Advanced audit views (filter by user, date, entity type)	Medium	10	×1.5	**15**	Suggested enhancement
45	Admin	System configuration UI (thresholds, settings — avoid hardcoding)	Medium	14	×1.5	**21**	Suggested enhancement
46	QA	UAT support + bug fixes from user feedback	Medium	24	×1.5	**36**	Standard
					Post-MVP Total	**~117 hrs**	

Backlog (Could Have)

#	Task	Source
47	Dashboard with key metrics (active offers, pending approvals, recent bids)	Suggested enhancement
48	Bulk bid operations (approve/reject multiple at once)	Suggested enhancement
49	Advanced search and filtering across all entities	Suggested enhancement
50	Email notifications on approval status changes	Suggested enhancement
51	Data export API for future external consumption	Suggested enhancement
52	User activity log (who logged in when)	Suggested enhancement

Out of Scope (Won’t Have)

- Customer-facing portal (future phase)
- In-app analytics dashboards (future consideration)
- Commodities pricing database / AI training data
- Mobile application
- Integration with external ERP or order management systems

Learning Curve Budget

Technology	Who	Hours	Justification
Blazor component model + render modes	Dev 1	12	Components, lifecycle, SSR vs Interactive
Blazor component model + render modes	Dev 2	12	Same
MudBlazor (DataGrid + UI components)	Dev 1	8	Grid API, cell editing, forms
MudBlazor (DataGrid + UI components)	Dev 2	6	Support knowledge
Microsoft.Identity.Web (Blazor auth)	Dev 1	6	Built-in auth patterns
Azure App Service + SQL (if new)	Both	8	Deployment, connection strings, config
CI/CD pipeline setup	TBD	4	Single project = simpler
	Learning Curve Total	**56 hrs**	

Summary

Concept	Hours
MVP (Must Have) — Auth Option A: Azure AD Groups	~990 hrs
Post-MVP (Should Have)	~117 hrs
Learning Curve	~56 hrs
Full Scope (MVP + Post-MVP + Learning)	~1,163 hrs

Updated Feb 2026 after client Q&A. Items added: inventory identity/dedupe (+61 hrs), bid uniqueness (+15 hrs), allocation reversal (+52 hrs), tie-breaker (+12 hrs), below floor indicator (+12 hrs). Thresholds simplified (-9 hrs). 27 discovery questions still open.

Timeline Scenarios — Auth Option A (2 developers, 80 hrs/week)

Scenario	Scope	Hours	Working Weeks	Calendar
MVP only	Must Have + Learning	~1,046 hrs	~13 weeks	~3.25 months
MVP + Post-MVP	Full Phase 1	~1,163 hrs	~14.5 weeks	~3.6 months

With 3 developers (120 hrs/week)

Scenario	Scope	Hours	Working Weeks	Calendar
MVP only	Must Have + Learning	~1,046 hrs	~8.5 weeks	~2.1 months
MVP + Post-MVP	Full Phase 1	~1,163 hrs	~9.7 weeks	~2.4 months

Auth Option B adds ~24 hrs. Timelines assume productive from week 1 after learning curve. Ramp-up may add 1-2 weeks. Does not include UAT.

Risk Assessment

Risk	Probability	Impact	Hours at Risk	Mitigation
MudBlazor DataGrid insufficient for in-cell editing	Medium	High	+16-24 hrs + $1K/dev	Evaluate in Sprint 0; Telerik/Syncfusion as fallback
SignalR connection reliability	Low	Medium	+8-16 hrs	Built-in reconnection; test under real network
Blazor community smaller — harder to find solutions	Medium	Medium	+8-16 hrs per incident	MudBlazor Discord, Microsoft docs
Partial/split allocation business rules unclear	High	High	+24-40 hrs	Define rules with Product Owners before coding
Allocation reversal edge cases (partial reversals, already-exported data)	Medium	High	+16-24 hrs	Define reversal rules explicitly with POs; test with real scenarios
Inventory composite uniqueness conflicts	Medium	Medium	+8-16 hrs	Define exact uniqueness rules with client; validate during Sprint 0
CSV upload edge cases (bad data, large files)	Medium	Medium	+8-16 hrs	Define validation rules upfront; limit file sizes
Azure infra not available on time	Medium	High	Blocks 1-2 weeks	Confirm access before Sprint 1
Scope creep from 3 Product Owners	Medium	High	+40-80 hrs	Reference charter; change request process

TBD Impact on Estimation

Unknown	If Confirmed As...	Impact on Hours
Auth approach: Azure AD vs Internal DB	Option A: Azure AD Groups (recommended)	Auth = 68 hrs (baseline)
Auth approach: Azure AD vs Internal DB	Option B: Internal DB roles + admin UI	Auth = 92 hrs (+24 hrs)
3rd developer available	Yes	Timeline compresses ~30%
3rd developer available	No	Stays at 2-dev scenario
MudBlazor DataGrid sufficient	No — need Telerik/Syncfusion	+$1-2K license, +8-16 hrs
Automated infrastructure setup	Yes	+16-24 hrs

Team Sign-off

Name	Role	Approved
TBD	Simpat Lead	
TBD	Dev 1	
TBD	Dev 2	